

Formulae

General & Properties

$$h = u + Pv$$

$$y_x = y_f + xy_{fg}$$

$$x = \frac{m_{\text{sat. vapour}}}{m_{\text{total}}} = \frac{m_{\text{sat. vapour}}}{m_{\text{sat. liquid}} + m_{\text{sat. vapour}}}$$

Ideal Gas

$$Pv = RT$$

$$PV = mRT$$

$$PV = NR_uT$$

$$\frac{P_1V_1}{T_1} = \frac{P_2V_2}{T_2}$$

$$\Delta u = C_v \Delta T$$

$$\Delta U = m C_v \Delta T$$

$$C_p = C_v + R$$

$$R_u = 8.314 \frac{\text{kJ}}{\text{kmol K}}$$

$$\Delta h = C_p \Delta T$$

$$\Delta H = m C_p \Delta T$$

$$\gamma = k = \frac{C_p}{C_v}$$

$$R = \frac{R_u}{M}$$

First Law – Closed Systems

$$W_{b, \text{out}} = \int_1^2 P dV$$

$$\underbrace{E_{\text{in}} - E_{\text{out}}}_{\text{net energy transfer by heat and work}} = \underbrace{\Delta E_{\text{system}}}_{\text{change in internal, kinetic potential, etc energies}}$$

$$Q - W = \Delta E_{\text{system}} = \Delta U$$

First Law – Open Systems

$$\sum \dot{m}_{\text{in}} - \sum \dot{m}_{\text{out}} = \Delta \dot{m}_{\text{system}}$$

$$\dot{m} = \rho v A = \rho \dot{V} = \frac{\dot{V}}{v}$$

$$\underbrace{\dot{E}_{\text{in}} - \dot{E}_{\text{out}}}_{\text{rate of net energy transfer by heat, work and mass}} = \underbrace{\Delta \dot{E}_{\text{system}}}_{\text{rate of change in internal, kinetic, potential, etc energies}}$$

Steady Flow Processes

$$\sum \dot{m}_{\text{in}} = \sum \dot{m}_{\text{out}}$$

$$\rho_1 v_1 A_1 = \rho_2 v_2 A_2$$

$$\dot{Q} - \dot{W} = \underbrace{\sum \dot{m}_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right)}_{\text{for each outlet}} - \underbrace{\sum \dot{m}_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right)}_{\text{for each inlet}}$$

$$\dot{Q} - \dot{W} = \dot{m} [\Delta h + \Delta e_k + \Delta e_p] = \dot{m} \left[h_2 - h_1 + \frac{v_2^2 - v_1^2}{2} + g(z_2 - z_1) \right]$$

Uniform Flow Processes

$$\sum m_{\text{in}} - \sum m_{\text{out}} = (m_{\text{final}} - m_{\text{initial}})_{\text{system}}$$

$$Q - W = \sum m_{\text{out}} \left(h_{\text{out}} + \frac{v_{\text{out}}^2}{2} + gz_{\text{out}} \right) - \sum m_{\text{in}} \left(h_{\text{in}} + \frac{v_{\text{in}}^2}{2} + gz_{\text{in}} \right) + (m_2 e_2 - m_1 e_1)_{\text{system}}$$

$$Q - W = \sum m_{\text{out}} h_{\text{out}} - \sum m_{\text{in}} h_{\text{in}} + (m_2 u_2 - m_1 u_1)_{\text{system}}$$

Second Law- Heat Engines, Refrigerators and Heat Pumps

$$\eta_{\text{th}} = \frac{W_{\text{net, out}}}{Q_{\text{H}}} = 1 - \frac{Q_{\text{L}}}{Q_{\text{H}}}$$

$$\eta_{\text{th, rev}} = 1 - \frac{T_{\text{L}}}{T_{\text{H}}}$$

$$\text{COP}_{\text{R}} = \frac{Q_{\text{L}}}{W_{\text{net, in}}}$$

$$\text{COP}_{\text{R, rev}} = \frac{1}{\frac{T_{\text{H}}}{T_{\text{L}}} - 1}$$

$$\text{COP}_{\text{H}} = \frac{Q_{\text{H}}}{W_{\text{net, in}}}$$

$$\text{COP}_{\text{H, rev}} = \frac{1}{1 - \frac{T_{\text{L}}}{T_{\text{H}}}}$$

Entropy

$$\oint \frac{\delta Q_{\text{net}}}{T} \leq 0$$

$$dS = \left(\frac{\delta Q}{T} \right)_{\text{int rev}}$$

$$(S_{\text{system}} + S_{\text{surrounding}}) = S_{\text{gen}} \geq 0$$

$$T ds = du + P dv$$

$$T ds = dh - v dP$$

Entropy Changes and Isentropic Relations

Incompressible substances:

$$s_2 - s_1 = C_{\text{avg}} \ln \frac{T_2}{T_1}$$

$$\text{isentropic process: } T_2 = T_1$$

Ideal Gases – constant specific heats

$$s_2 - s_1 = C_{\text{v, avg}} \ln \frac{T_2}{T_1} + R \ln \frac{v_2}{v_1}$$

$$s_2 - s_1 = C_{\text{p, avg}} \ln \frac{T_2}{T_1} - R \ln \frac{P_2}{P_1}$$

$$\bar{s}_2 - \bar{s}_1 = \bar{C}_{\text{v, avg}} \ln \frac{T_2}{T_1} + R_{\text{u}} \ln \frac{v_2}{v_1}$$

$$\bar{s}_2 - \bar{s}_1 = \bar{C}_{\text{p, avg}} \ln \frac{T_2}{T_1} - R_{\text{u}} \ln \frac{P_2}{P_1}$$

Ideal Gases – variable specific heats

$$s_2 - s_1 = s_2^{\circ} - s_1^{\circ} - R \ln \frac{P_2}{P_1} \quad (\text{kJ/kg.K})$$

$$\bar{s}_2 - \bar{s}_1 = \bar{s}_2^{\circ} - \bar{s}_1^{\circ} - R_{\text{u}} \ln \frac{P_2}{P_1} \quad (\text{kJ/kmol.K})$$

Isentropic Process – constant specific heats

$$\left(\frac{T_2}{T_1} \right)_{s=\text{const}} = \left(\frac{P_2}{P_1} \right)^{\frac{k-1}{k}} = \left(\frac{v_1}{v_2} \right)^{k-1}$$